

Business Overview

Introduction to B2B Operations:

The B2B world operates differently from the B2C or C2C world. Businesses work with other businesses on **credit**. When a **buyer business** orders goods from the **seller business**, the **seller business issues an invoice for the same**. This invoice for the goods contains various information like the details of the goods purchased and when it should be paid. This is known in accounting terminology as “Accounts Receivable”

“Accounts Receivable represents money owed by entities to the firm on the sale of products or services on credit. In most business entities, accounts receivable is typically executed by generating an invoice and either mailing or electronically delivering it to the customer, who, in turn, must pay it within an established timeframe, called credit terms or payment terms.”

Seller business interacts with various businesses and sells goods to all of them at various times. Hence, the seller business needs to keep track of the total amount it owes from all the buyers. This involves keeping track of all invoices from all the buyers. Each invoice will have various important fields like a payment due date, invoice date, invoice amount, baseline date etc.

The buyer business needs to clear its amount due before the due date. However, in real-world scenarios, the invoices are not always cleared ie. paid in full amount by the due date. The date on which a customer clears the payment for an invoice is called the **payment date**.

Account receivables Department:

1. In the ideal world, the buyer business should payback within the stipulated time (ie the Payment Term). However, in the real world, the buyer business seldom pays within their established time frame, and this is where the Account receivables Department comes into picture.
2. Every business consists of a dedicated Account receivables Department to collect and track payment of invoices.
3. It consists of a Account receivables team that is responsible for:
 - Collecting payments from customers for their past due invoices
 - Sending reminders and follow ups to the customers for payments to be made
 - Looking after the entire process of getting the cash inflow
 - Help the company get paid for the services and products supplied.

Problem Statement for Application Development:

- Data Preprocessing of invoices dataset
- Feature Engineering on the dataset
- To build a Machine Learning Model to predict the payment date of an invoice when it gets created in the system

FUNCTIONAL OVERVIEW

Milestone 1: Data Preprocessing

We will be providing you an **invoices dataset** which you need to parse and process.

account_id	document_numbr	company_code	fiscal_year	branch	customer_number	fk_customer_ma	document_date	baseline_date_n	due_date
60	20000950	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000950	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000950	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000950	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000870	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000870	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000870	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000870	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20001004	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20001004	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20001004	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000831	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000831	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000831	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20000831	IN	IN	IN	999888	-1	05-04-16	IN	IN
60	20103019	IN	IN	IN	999888	-1	11-04-16	IN	IN
60	20103019	IN	IN	IN	999888	-1	11-04-16	IN	IN
60	20103019	IN	IN	IN	999888	-1	11-04-16	IN	IN
60	20000677	IN	IN	IN	999888	-1	05-04-16	IN	IN

List of all the fields part of dataset are as follows:

- Acct Doc ID
- Company ID
- Document Number
- Document Number Norm
- Business Code

- Create Year
- Document type
- Customer Number
- Customer Number Norm
- Customer Map ID
- Customer Name
- Document Create Date
- Document Create Date Norm
- Posting date
- Posting date norm
- Posting ID
- Due In Date
- Due In Date Norm
- Order Create Date
- Order Create Date Norm
- Invoice ID
- Invoice ID Norm
- Baseline Create Date
- Invoice Date norm
- Total Open Amount
- Customer Payment Terms
- Shipping date
- Shipping to
- Clear Date
- Clear Date Norm
- Is Open Invoice
- Doc Id
- Actual Outstanding amount
- Invoice Amount
- Dispute Valid Status
- Dispute Amount

Milestone 2: Feature Engineering

Objective of this milestone is to apply feature engineering techniques to extract patterns and identify features for efficient modelling.

Milestone 3: Developing ML Model

An Invoices dataset that contains the past payment information and behaviour of various buyers. Based on the previous payment patterns, the ML model needs to predict what will be the date a payment is made by the customer for an invoice.

The model also needs to predict which aging bucket the invoice falls into based on the predicted payment date.

For example:

<u>Invoice No</u>	<u>Due Date</u>	<u>Actual Open Amt(\$)</u>	<u>Payment Date</u>
1001	03/12/2020	9000	14/02/2021
1002	10/12/2020	5000	23/12/2020
1003	17/12/2020	3000	16/01/2021

In this example, the payment for the first invoice (1001) is due on 3rd December 2020. The payment for the invoice was actually made on 14th February 2021 so it was 73 days past due. Similarly for the second invoice(1002) which was due on 10th December 2020, the payment was actually made on 23rd December 2020, 13 days after the original due date. So the objective is to build a model predict the payment date for each of the invoices.